

Q1

Considering a computer system with a 10-bit logical address. Translate the logical address 1000011011 to the corresponding physical address for the following two cases:

- a) Assuming paging is used, the page size is 128 bytes and the page table is as given in Figure Q4a.
- b) Assuming segmentation is used, the maximum segment size that the system can support is 256 bytes, and the segment table is as given in Figure Q4b.

0	01011
1	00010
2	00010
3	10101
4	01001
5	01100
6	11010
7	00110

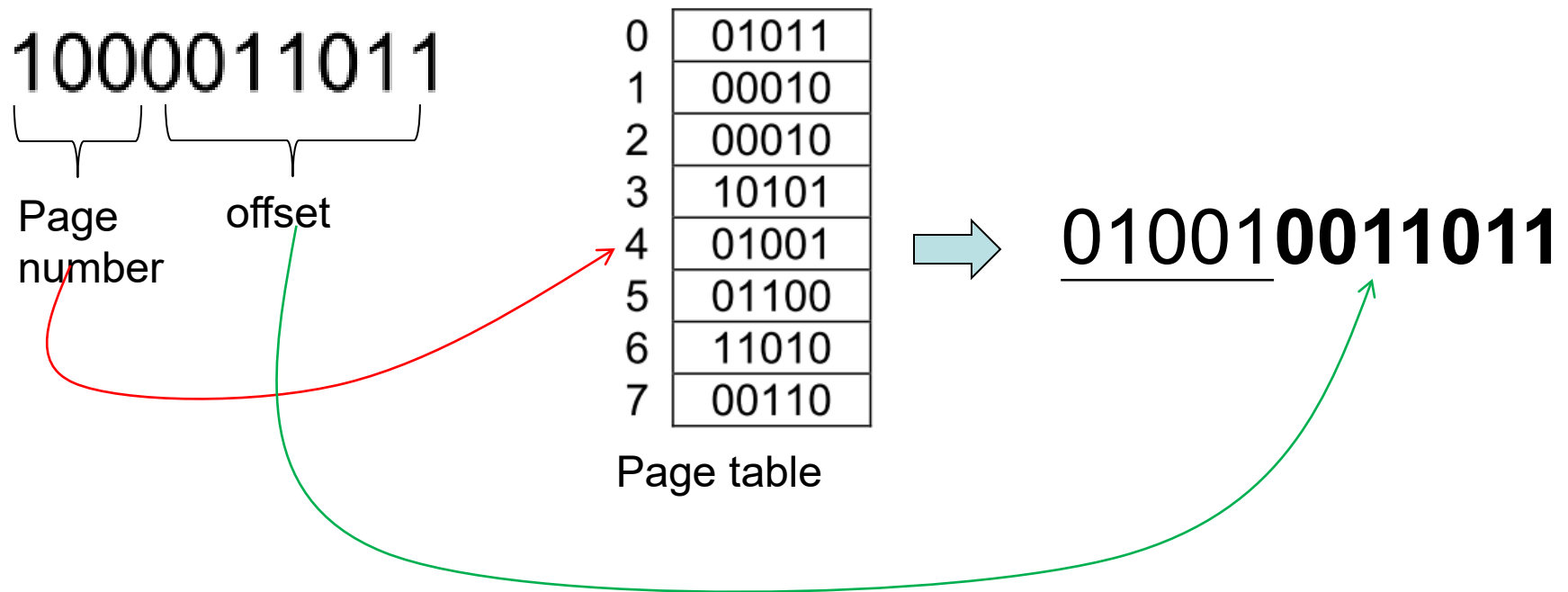
Figure Q4a

0	00010000	010000000000
1	00001000	100000000000
2	00100000	010011010000
3	00011000	100001100000

Figure Q4b

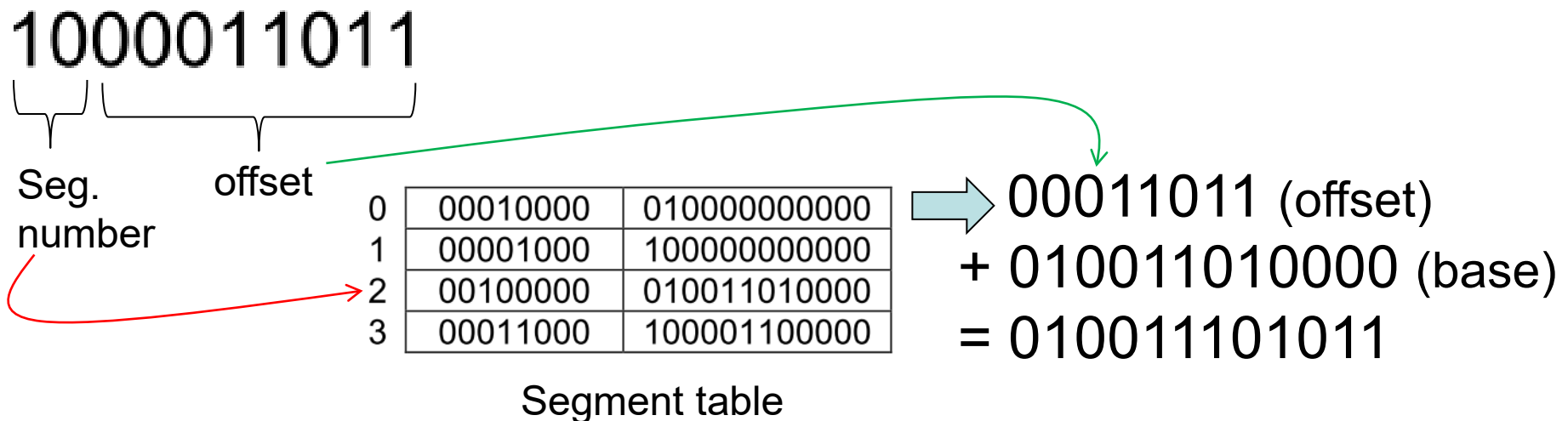
Q1 (a)

- Given page size is 128 bytes → 7 bits are needed for page offset and 3 bits for page number.



Q1 (b)

- The maximum segment size is 256 bytes → 8 bits are needed for the segment offset and 2 bits for segment number.



Q2

A paged memory system uses the page size of 1024 bytes. Size of a page table entry is 4 bytes and the logical address space is 2^{30} bytes.

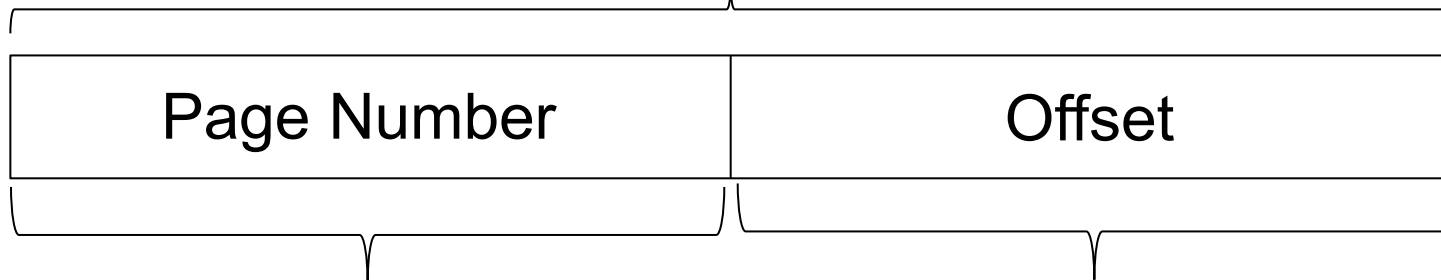
- a) What is the size of the page table if single level of paging is used?
- b) What is the minimum number of levels of page tables needed in this system to ensure that the outmost page table will fit within a single page frame?
- c) Draw an address translation diagram to show how logical address translation is performed.

Q2 (Answer)

- a) size of page table: $\#pages \times page_table_entry_size$
 $= 2^{30}/2^{10} \times 4 = 2^{22} = 4 \text{ megabytes}$
- b) If two-level paging is used, size of outer page table: $2^{22}/2^{10} \times 4 = 2^{14} = 16k \text{ bytes} > 1024 \text{ bytes}$.
If three-level paging is used, size of outer page table: $2^{14}/2^{10} \times 4 = 2^6 \text{ bytes} < 1024 \text{ bytes}$.
→ at least three levels.

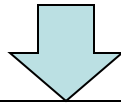
Q2 (c)

30 bits



of Pages
 $30 - 10 = 20$ bits

Page size
 $1024 \text{ bytes} \rightarrow 10 \text{ bits}$



P1	P2	P3
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Hints:

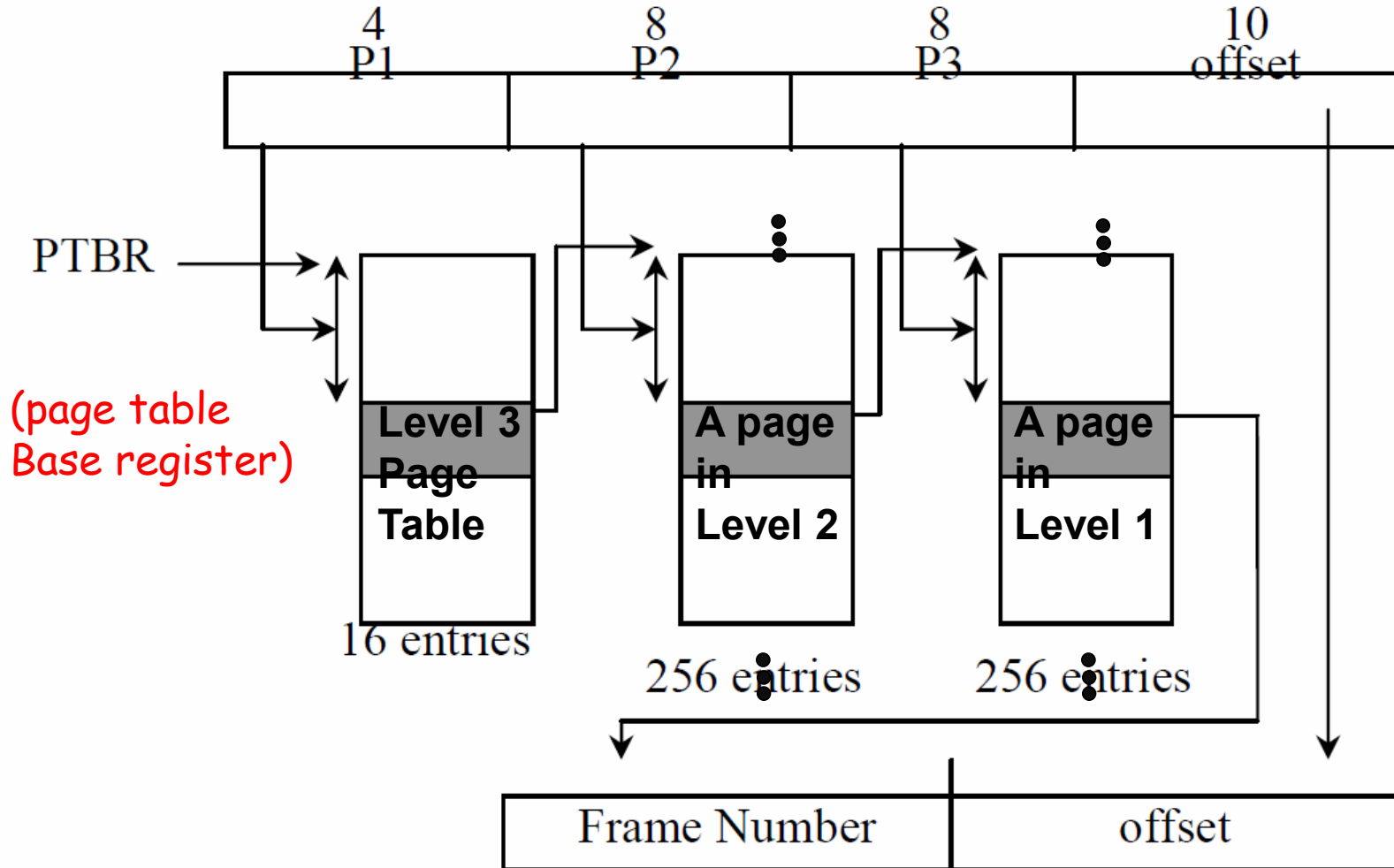
P1: the index in the level 3 (outmost). Each entry in level 3 page table points to a page in Level 2.

P2: the index within a page in the level 2. Each entry in level 2 page table points to a page in Level 1.

P3: the index within a page in the level 1. Each entry in level 1 page table points to a memory frame.

Q2 (c)

- c) Each page can contain $2^{10}/2^2 = 2^8$ entries, the outmost page table has 2^4 entries. The virtual address format is as shown in the following figure:



Q3

1. A computer has four page frames. The time of loading, time of last access, and the R bit for each page are as shown below (the times are in clock ticks):

<u>PAGE</u>	<u>LOADED</u>	<u>LAST ACCESS.</u>	<u>R</u>
0	126	279	0
1	230	260	0
2	120	272	1
3	160	280	1

Q3 (a)

a) Which page will FIFO replace?

<u>PAGE</u>	<u>LOADED</u>	<u>LAST ACCESS.</u>	<u>R</u>
0	126	279	0
1	230	260	0
2	120	272	1
3	160	280	1

Q3 (b)

b) Which page will second chance replace?

(different from FIFO)

PAGE	LOADED	LAST ACCESS.	R
0	126	279	0
1	230	260	0
2	120	272	1
3	160	280	1

Q3 (c)

c) Which page will LRU replace?

<u>PAGE</u>	<u>LOADED</u>	<u>LAST ACCESS.</u>	<u>R</u>
0	126	279	0
1	230	260	0
2	120	272	1
3	160	280	1

Q4

2. For each of the page replacement policies listed below, calculate the number of page faults encountered when referencing the following pages. Assume the availability of 4 empty page frames.

0 1 6 0 3 4 0 1 0 3 4 6 3 4

a) FIFO

b) CLOCK

c) LRU

Q4 (FIFO)

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0	0	0	0	0	4	4	4	4	4	4	4	3	3
F2		1	1	1	1	1	0	0	0	0	0	0	0	4
F3			6	6	6	6	6	1	1	1	1	1	1	1
F4					3	3	3	3	3	3	3	6	6	6
						0	1	6				3	4	0
	P	P	P		P	P	P	P				P	P	P

There are 10 page faults.

Q4 (CLOCK)

[illegible]

There are 7 pages faults.

Clock hand in shaded entry.

Q4 (LRU)

	0	1	6	0	3	4	0	1	0	3	4	6	3	4
F1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
F2		1	1	1	1	4	4	4	4	4	4	4	4	4
F3			6	6	6	6	6	1	1	1	1	6	6	6
F4					3	3	3	3	3	3	3	3	3	3
						1		6				1		
	P	P	P		P	P		P				P		

There are 7 page faults.